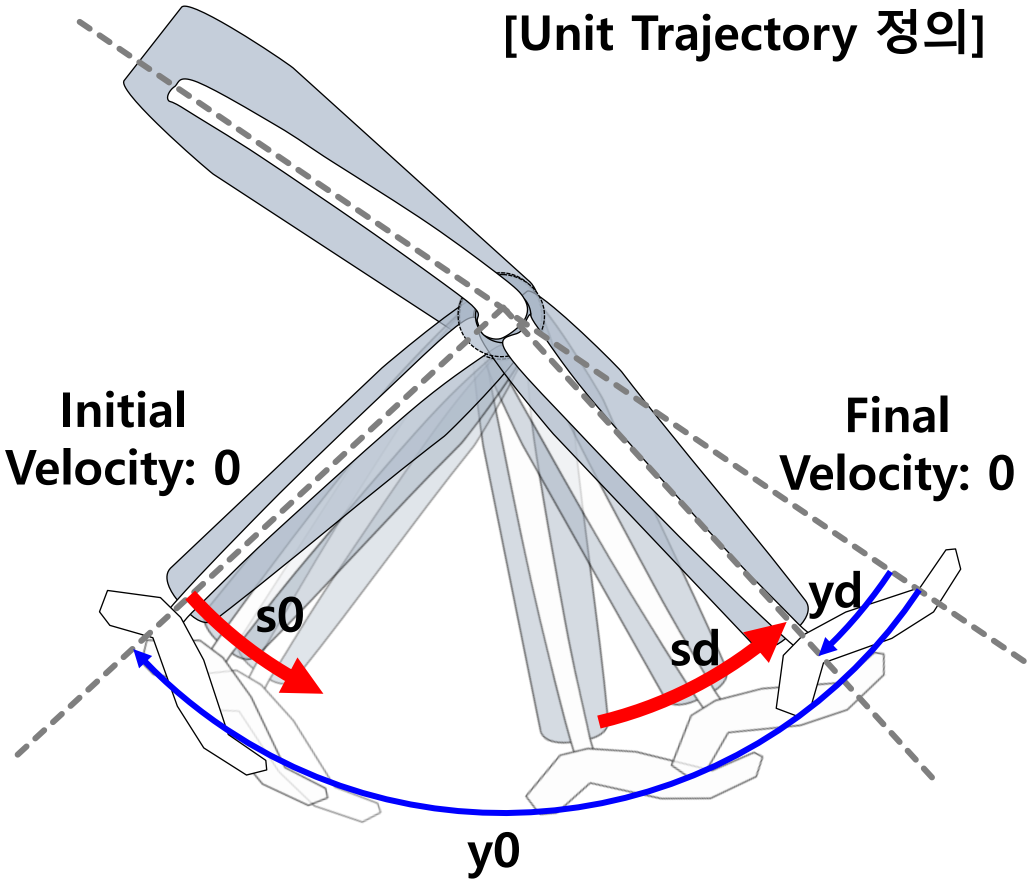


정지된 초기위치 y0에서, s0만큼 가속하면서 이동을 시작하고, 다시 sd 이만큼 감속하면서 목적 위치 yd 까지 이동

y0 : 초기 위치

- Physical unit : degree (Output축 기준)
- Range : -360 ~ 360
- Neutral posture : 0
- Data format : int16

yd : 목적 위치

- Unit and format : the same as y0

s0 : 가속 파라미터

- Physical unit : 무차원수
- Range : -128 ~ 128
- Data format : int8

sd : 감속 파라미터

- Unit and format : the same as s0

A unit trajectory, representing a unit movement, is expressed by a **5th-order high-order polynomial** as shown below

$$y(\tau) = a_0 + a_2\tau^2 + a_3\tau^3 + a_4\tau^4 + a_5\tau^5$$

Here, each variable and coefficient is defined as follows

$$\tau = \frac{k}{L_{trajectory}} \in [0,1], \qquad k \in [0,1,2\ldots, L_{trajectory}]$$

$$a_0 = y_0$$

$$a_2 = 0.5s_0(y_d - y_0)$$

$$a_3 = (10 - 1.5s_0 + 0.5s_d)(y_d - y_0)$$

$$a_4 = (-15 + 1.5s_0 - s_d)(y_d - y_0)$$

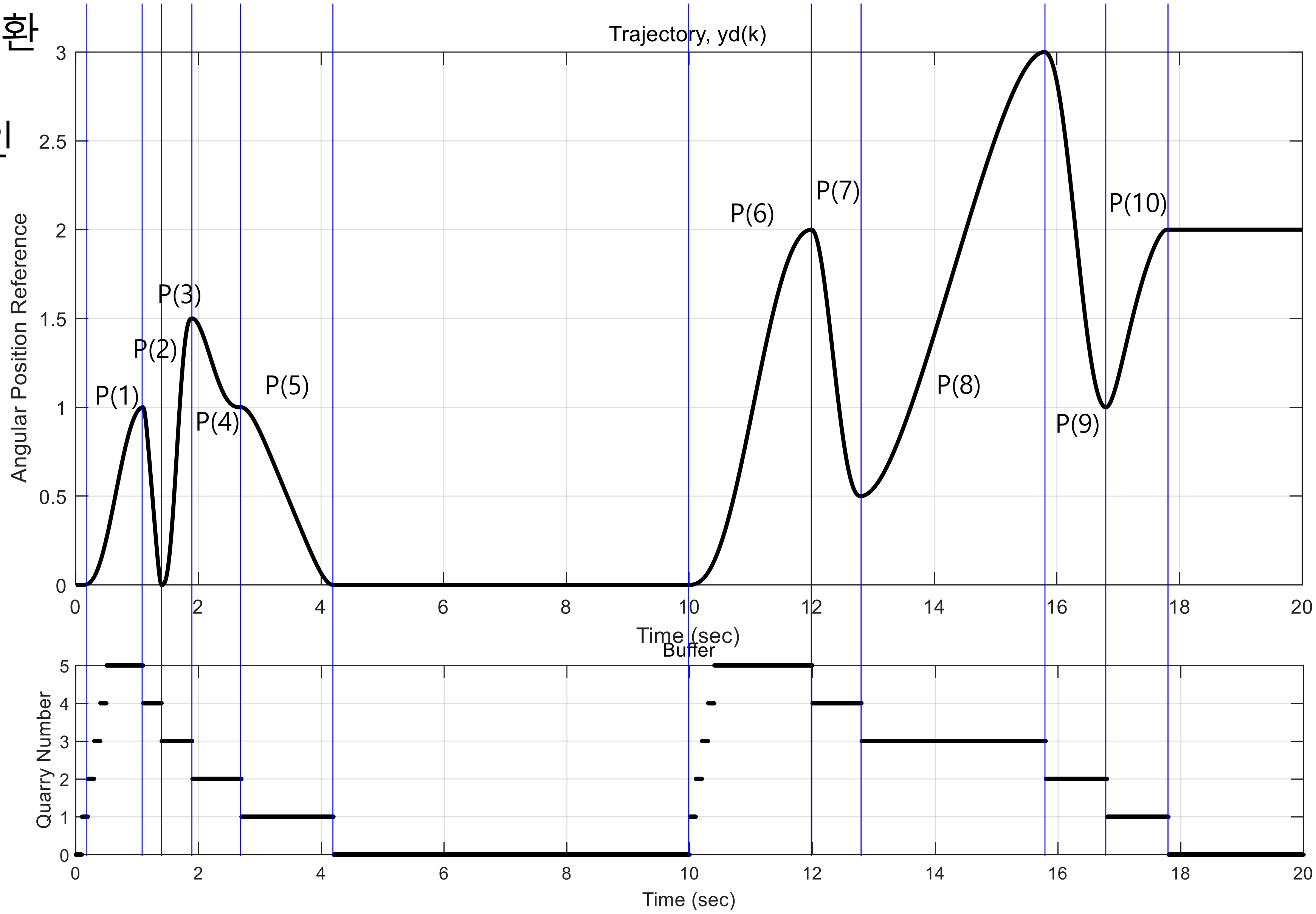
$$a_5 = (6 - 0.5s_0 + 0.5s_d)(y_d - y_0)$$

즉, 주어진 y_0 에 대해 단위 동작은 y_d, s_0, s_d, $L_{trajectory}$의 함수
→ P(i) = [yd, Ltraj, s0, sd] 파라미터 벡터로 단위 궤적 정의 가능

P-Vector: 파라미터 기반 Position 궤적 생성

- 촘촘한 픽셀 → 벡터로 변환
- SD카드 접근 → [MotionMap.csv] File 확인

P(1) =	[200	1000	0	5];
P(2) =	[0	300	10	10];
P(3) =	[300	500	0	0];
P(4) =	[200	800	5	0];
P(5) =	[0	1500	10	10];
P(6) =	[400	2000	3	8];
P(7) =	[100	800	10	3];
P(8) =	[600	3000	20	20];
P(9) =	[200	1000	7	8];
P(10)=	[400	1000	12	6];
	↓ y_d	↓ L_{traj} (궤적길이)	↓ s_0	↓ s_d



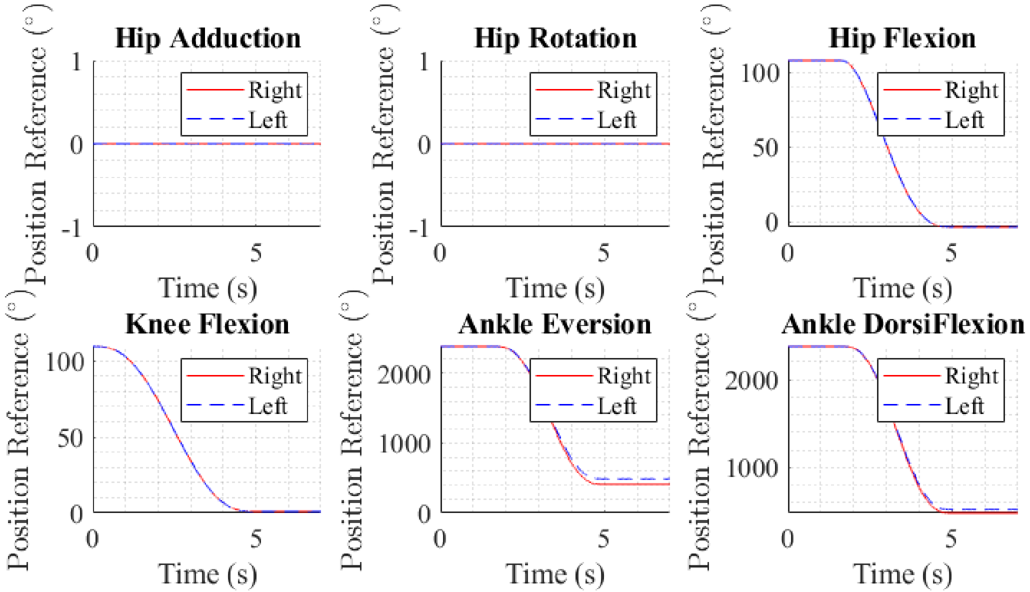
Motion Map (예시 : 일어서기)

MotionSet ID: *SIT2STAND*

"MotionMap.csv" file

MS ID	MS NAME	MD ID	C-Vector	P-Vector		
				1	2	
5	SIT2STAND	MD1	...	0, 5000, 0, 0	-	...
		MD2	...	0, 5000, 0, 0	-	...
		MD3	...	0, 5000, 0, 0	-	...
		MD4	...	0, 5000, 0, 0	-	...
		MD5	...	108, 1500, 0, 0	-3.6, 3333, 10, 0	...
		MD6	...	108, 1500, 0, 0	-3.6, 3333, 10, 0	...
		MD7	...	1, 5000, 0, 0	-	...
		MD8	...	1, 5000, 0, 0	-	...
		MD9	...	2389.4,1500.0,0.0,0.0	448.5446,3333.0,0.0,0.0	...
		MD10	...	2444.8,1500.0,0.0,0.0	531.2969,3333.0,0.0,0.0	...
		MD11	...	2396.3,1500.0,0.0,0.0	524.0605,3333.0,0.0,0.0	...
		MD12	...	2440.0,1500.0,0.0,0.0	567.2713,3333.0,0.0,0.0	...

Decoding in MD



Motion Map (예시 : 걷기)

MotionSet ID: *NORMAL_WALKING*

“MotionMap.csv” file

MS ID	MS NAME	MD ID	C-Vector	P-Vector		
				1	2	
28	NORMAL_WALKING	MD1	...	322.2, 182.0, 13.8, 5.5	126.3, 390.0, 18.4, 1.2	...
		MD2	...	55.4, 184.0, 13.1, 5.9	207.2, 268.0, 8.0, 0.0	...
		MD3	...	0.5, 1020.0, 0.0, 0.0	-	...
		MD4	...	0.5, 1020.0, 2.5, 102.5	-	...
		MD5	...	45.7, 567.0, 3.7, 8.0	30.8, 453.0, 16.4, 0.0	...
		MD6	...	-2.7, 1020.0, 23.6, 28.8	-	...
		MD7	...	73.9, 488.0, 11.2, 7.3	23.4, 360.0, 4.3, 2.5	...
		MD8	...	9.0, 1020.0, 23.2, 27.5	-	...
		MD9	...	1375.2, 341.0, 0.5, 7.8	2072.3, 256.0, 9.7, 1.0	...
		MD10	...	1621.1, 387.0, 23.4, 3.5	1890.2, 308.0, 1.1, 4.1	...
		MD11	...	1392.8, 350.0, 1.3, 9.8	2034.5, 300.0, 15.3, 0.0	...
		MD12	...	2023.6, 447.0, 14.5, 0.3	2717.0, 573.0, 14.9, 33.1	...

Decoding in MD